Week 3 Network Security Monitor

Abstract

This project implements a Network Security Monitoring System using Python. It captures live network packets using the Scapy library and stores them in a SQLite database. The system detects suspicious activities such as port scanning, packet flooding, and SYN flood attacks. It also generates real-time alerts and provides both console-based and GUI-based monitoring interfaces.

Introduction

Network security is crucial to protect systems from cyber threats. Monitoring network traffic helps in identifying malicious activities early. This project focuses on building a network monitoring tool that captures packets, analyzes traffic patterns, and detects suspicious behaviors like attacks or unusual access attempts.Tools Used

- Python
- Scapy (Packet Capture)
- SQLite (Database)
- Tkinter (GUI)
- Pandas (Data Analysis)
- Threading

Steps Involved in Building the Project

1. Requirement Analysis
2. Packet Capturing using Scapy
3. Database Design using SQLite
4. Implementing Attack Detection (Port Scan, Flooding, SYN Attack)
5. Alert Generation System
6. GUI Development using Tkinter
7. Data Visualization using Pandas
8. Testing and Debugging

Conclusion

The Network Security Monitor successfully detects suspicious network activities and provides real-time alerts. It enhances understanding of cybersecurity concepts such as intrusion detection and traffic analysis. This project can be extended with advanced features like machine learning-based threat detection and cloud monitoring.